DR. ALEX J. COOPER, FHEA

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RESEARCH POSITIONS

Trottier Space Institute Postdoctoral Fellow *Aug 2026* - present
Trottier Space Institute, McGill University, Canada

Hintze Prize Postdoctoral Fellow *Apr 2023* - Aug 2026
University of Oxford, UK

PhD Researcher *Oct 2018 - Oct 2022*
University of Amsterdam, Netherlands

EDUCATION

PhD Astrophysics *Oct 2018 - Oct 2022*
University of Amsterdam
Thesis: Multi-messenger and Multi-wavelength Signatures of Compact Object Transients
Supervisors: Prof. Ralph Wijers & Dr. Antonia Rowlinson

MSc Astrophysics (Gravitation and Astroparticle track) *Sep 2016 - Sep 2018*
University of Amsterdam & Vrije Universiteit Amsterdam
Thesis: Cosmic Rays from X-ray Binary Jets
Supervisor: Prof. Sera Markoff

BSc First Class, Physics and Philosophy *Sep 2013 - Sep 2016*
University of Nottingham
Thesis: Object Permanence in Minkowski Space-time
Supervisor: Dr. Dave Ingram

CERTIFICATIONS

Fellow of the Higher Education Academy *Feb 2026*
AdvanceHE

RESEARCH INTERESTS

I am interested in modelling the physical processes powering emission from neutron stars and black holes, to better understand the extreme fundamental physics that governs them. My interests include:

- The sources and emission mechanisms of fast radio bursts and long period transients
- Multi-messenger/wavelength modelling of astrophysical jets
- Binary magnetospheric interaction of compact object mergers and star-planet systems
- The origin of high-energy cosmic rays and neutrinos

VISITING RESEARCH POSITIONS

Visiting Postdoctoral Fellow *Jan 2026 - Feb 2026*
T. D. Lee Institute, Shanghai, China

INVITED ORAL PRESENTATIONS

Invited Talk, FRB2025, Guiyang, China	07/2026
Colloquium, Shanghai Astronomical Observatory CAS, China	02/2026
Astrophysics Seminar, T D Lee Institute, China	01/2026
Colloquium, CSIRO Sydney, Australia	07/2025
Invited Overview Talk, FRB2025, Canada/Online	07/2025
Astrophysics Colloquium, University of Edinburgh, UK	04/2025
Astrophysics Colloquium, NASA Goddard Spaceflight Centre, USA	03/2025
Transient Discussion Seminar, McGill University, Canada	03/2025
Astrophysics Colloquium, University of Alicante, Spain	09/2024
Invited Conference Talk, 17th Marcel Grossman Meeting, Italy	07/2024
Invited Conference Talk, Dialogue at the Dreamfield, China	05/2024
SPIMAX Seminar, Oxford University, UK	05/2022
Astrophysics Seminar, Istanbul University Observatory Seminar, Online	04/2022
Astrophysics Lunch Seminar, The George Washington University, USA	01/2022
Black Hole Institute Seminar, Harvard University, USA	01/2022

SELECTED CONTRIBUTED PRESENTATIONS

Microquasars Workshop, Italy	09/2025
Dynamic Radio Sky, Australia	08/2025
University of Sydney (Radio Transients Group), Australia	07/2025
University of Leicester (High-energy Transients Group), UK	06/2025
SPINS-UK 2024, UK	09/2024
17th Marcel Grossman Meeting, Italy	07/2024
Fast Radio Bursts 2023, Online	11/2023
National Astronomical Meeting, UK	07/2023
Flatiron Institute (SPA Group), USA	01/2022
Fast Radio Bursts 2021 (Online)	08/2021
Nederlandse Astronomenclub, NL	06/2019

RESEARCH SUPERVISION

PhD Student Projects

- Emma Elley - Oxford — 01/2025 - present
  The role of the kink instability in astrophysical jets (Co-supervisor: J. Matthews)

MSc Students

- James Jakab - McGill — 09/2026 - present
  Modelling binary long period transients (Co-supervisor: J. Hessels)

- Jin Zheng - Oxford (now in industry) 09/2024 - 07/2025
  Identifying optical counterparts to IceCube neutrinos
- Nicky Bacon - Oxford (now PhD @ Cambridge) 09/2023 - 07/2024
  Extreme particle acceleration in discrete black hole ejections

Summer Students

- Ellen Brabban-Brown - Oxford (now PhD @ Leicester) 06/2025 - 08/2025
  Modelling self-absorbed flares from black holes
- Alana Swain - Oxford/Breakthrough Listen (now PhD @ Cambridge) 06/2024 - 08/2024
  Characterising coherent radio emission from star-planet interaction
- Isabel Stephens - Oxford (now in industry) 06/2024 - 08/2024
  Does Scorpius X-1 produce fast jets?
- Alex Scott - Oxford/UNIQ+ 06/2024 - 08/2024
  Deboosting relativistic jets from black holes and neutron stars
- Om Gupta - Amsterdam/ASPIRE (now PhD @ UT Austin) 06/2024 - 08/2024
  Electromagnetic interaction of highly magnetised binary systems

TEACHING (TOTALLING $\sim$ 70 HOURS)

- Lecturer/Tutor, C1 in MSc Physics, Oxford, UK 2023 - 2025
  Radiative Processes and High-energy Astrophysics for 2 years in Oxford tutorial system
- Teaching Assistant, Projects in BSc Astronomy, Amsterdam, NL 2018-2021
  Taught 6 groups of 3 students through project proposal, observations, analysis, and presentation
- Fellow of the Higher Education Academy 02/2026

SUCCESSFUL OBSERVING PROPOSALS AS PRINCIPAL INVESTIGATOR

- LOFAR - single cycle - 2 triggers (8.8 hours) - Accepted in 4 cycles, totalling 36 hours 2019-2022
  Rapid follow-up of magnetar flares: the search for coherent radio emission
- LOFAR – DDT - 4 hours 2020
  Follow-up of SGR 1935+2145 as it enters an active phase
- LOFAR – DDT - 4.4 hours 2021
  Low-frequency follow-up of SGR 1935+2154 as it enters a new phase
- LOFAR - DDT - 16 hours 2023
  The state transition of new candidate black hole X-ray binary Swift J1727.8-1603
- NICER - DDT - 10ks 2025
  The search for X-ray emission from long period transient CHIME J0630+25

SUCCESSFUL GRANTS AS PRINCIPAL INVESTIGATOR

- Trottier Space Institute Fellowship 2025
- Hintze Prize Fellowship 2022
- T. D. Lee Visiting Postdoctoral Fellow 2025
- Leids Kerkhoven-Bosscha Fonds - Conference funding - €500 2019
- Leids Kerkhoven-Bosscha Fonds - External visitor funding - €1000 2021

- Lockey fund - Conference funding - £1000 2023
- Lockey fund - Travel funding - £500 2023
- Christ Church Research Centre - Travel funding - £2500 2024
- Astor Grant - Travel funding - £1500 2024
- Christ Church Research Centre - Travel funding - £1700 2024
- Christ Church Research Centre - Travel funding - £4200 2025
- Christ Church Research Centre - Travel funding - £770 2026

PROFESSIONAL SERVICE

- FRB2026 Discussion Panel Co-lead (Emission Mechanisms)
- Black Holes 2026 Discussion Panel Co-lead (AI in Science)
- Expert Journal Reviewer: Nature, Nature Astro., Nature Comms., MNRAS, ApJ, A&A, JHEAP, & PRL 2020 - present
- Internal SKA Reviewer: Advancing Astrophysics with the SKA II 2025
- Reviewer for the University of Malta Internal Research Grants Program 2023
- Organiser for Oxford SPIMAX Seminar Series 2023 - present
- Member of the National Education Commission overseeing Dutch BSc-PhD astronomy 2019 - 2022
- Scientific Organising Committee of Dynamic Radio Sky Conference 2025
- Local Organising Committee of ThunderKAT Collaboration and BlackHolistic conferences 2024
- Organising Committee of the ASPIRE summer programme - Head of Admissions 2022

COLLABORATION MEMBERSHIP

- AstroFlash Collaboration - High cadence and resolution FRB observations 2026 - present
- SKA Transients Working Groups - Transients and Pulsars 2025 - present
- ATLAS Transients Collaboration - Extragalactic optical transients survey 2023 - present
- ENGRAVE Collaboration - Gravitational wave event follow-up with ESO facilities 2023 - present
- ThunderKAT/X-KAT Collaboration - MeerKAT observations of X-ray binary jets 2022 - present

FIRST AUTHOR JOURNAL PUBLICATIONS

1. *Observational biases and improved modelling of off-axis relativistic jets*
   Cooper, A. J., Scott, A. P.*, Rhodes, L., Carotenuto, F., Hughes, A. K., Matthews, J. H., Savard, K., Cowie, F. J., Elley, E. L., Lilje, C., Fender, R. (2026)
   MNRAS, 2026, 550, 1187 (* denotes undergraduate student)

2. *Joint radiative and kinematic modelling of X-ray binary ejecta: energy estimate and reverse shock detection*
   Cooper, A. J., Matthews, J. H., Carotenuto, F., Fender, R., Lamb, G. P., Russell, T. D., Sarin, N., Savard, K., Zdziarski, A. A. (2025)
   MNRAS, 2025, 541, 3518

3. *Beyond the Rotational Deathline: Radio emission from ultra-long period magnetars*
   Cooper, A. J. & Wadiasingh, Z. (2024)
   MNRAS, 2024, 533, 2133

4. *Pulsar Revival in Neutron Star Mergers: Multi-messenger prospects for the discovery of pre-merger coherent radio emission*
   Cooper, A. J., Gupta, O.*, Wadiasingh, Z., Wijers, R.A.M.J., Boersma, O., Andreoni, I., Rowlinson, A., Gourdji, K. (2023)
   MNRAS, 2023, 519, 3923

5. *Testing Afterglow Models of FRB 200428 with Early Post-burst Observations of SGR 1935+2154*
   Cooper, A. J., Rowlinson, A., Wijers, R. A. M. J., Bassa, C., Gourdji, K., Hessels, J., van der Horst, A., Kondratiev, V., Michilli, D., Pleunis, Z., Shimwell, T., ter Veen, S. (2022)
   MNRAS, 2022, 517, 5483

6. *Constraining the nature of FRB-emitting bunches via photo-magnetic cascades*
   Cooper, A. J. & Wijers, R.A.M.J. (2021)
   Submitted to MNRAS, arXiv:2110.15244.

7. *Coherent Curvature Radiation: Maximum luminosity and high-energy emission*
   Cooper, A. J. & Wijers, R. A. M. J. (2021)
   MNRAS Letters, 2021, 508, L32

8. *High-energy Cosmic Rays from X-ray Binary Jets*
   Cooper, A. J., Gaggero, D., Markoff, S., Zhang, S. (2020)
   MNRAS, 2020, 493, 3212